

Repair Request Data Dictionary

Data / Index Consistency + Performance Evidence Clarification

Document Control	Value
Document ID	RR-DD-001
Version	1.4 - Performance Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Supersedes	RR-DD-001 v1.3
Revision Scope	Correct cross-document performance field ambiguity and classify performance test data without adding business fields.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Repair Request Data Dictionary

พจนานุกรมข้อมูลระบบ Repair Request / MVP Baseline Revision

Document Control	Value
Document ID	RR-DD-001
Version	1.3 - Baseline Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	Requirement v1.5; BR v1.5; State v1.5; Use Case v1.6
Audience	SA, Developer, Tester, Database Architect, API Architect
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด canonical Entity, Field, datatype, mandatory, validation, security, audit and traceability.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. Objective and Baseline

Canonical entity/field dictionary for the revised MVP. Datatypes shown are logical unless SQL mapping is explicitly named. Identity mapping: **created_by = Requester**; **tenant_id = Customer/Company scope**. No client-supplied tenant/actor is authoritative.

2. Data Conventions

Area	Baseline
Keys	Transactional PK = UUID; business numbers unique/immutable.
Tenant/Security	tenant_id on transactional/control data; derive tenant/actor/scope from authenticated context.
Concurrency	Mutable aggregates use row_version; state-changing API validates current state + token.
Time	Store UTC; Site time_zone_id used only for display; audit time immutable.
Audit	Append-only entity/id/action/actor/time/from/to/old/new/reason/correlation.
Files	Allowlist + <=10MB + SHA-256 + private ref + scan; only CLEAN satisfies evidence.
Retention	Organization policy / deployment config; business API cannot physical-delete Audit History.
Master Data	Priority, Outcome, Evidence Type, Category, Site, Equipment, Location, Team, Approval Route active; not hardcoded.

3. Entity Relationship / Ownership Summary

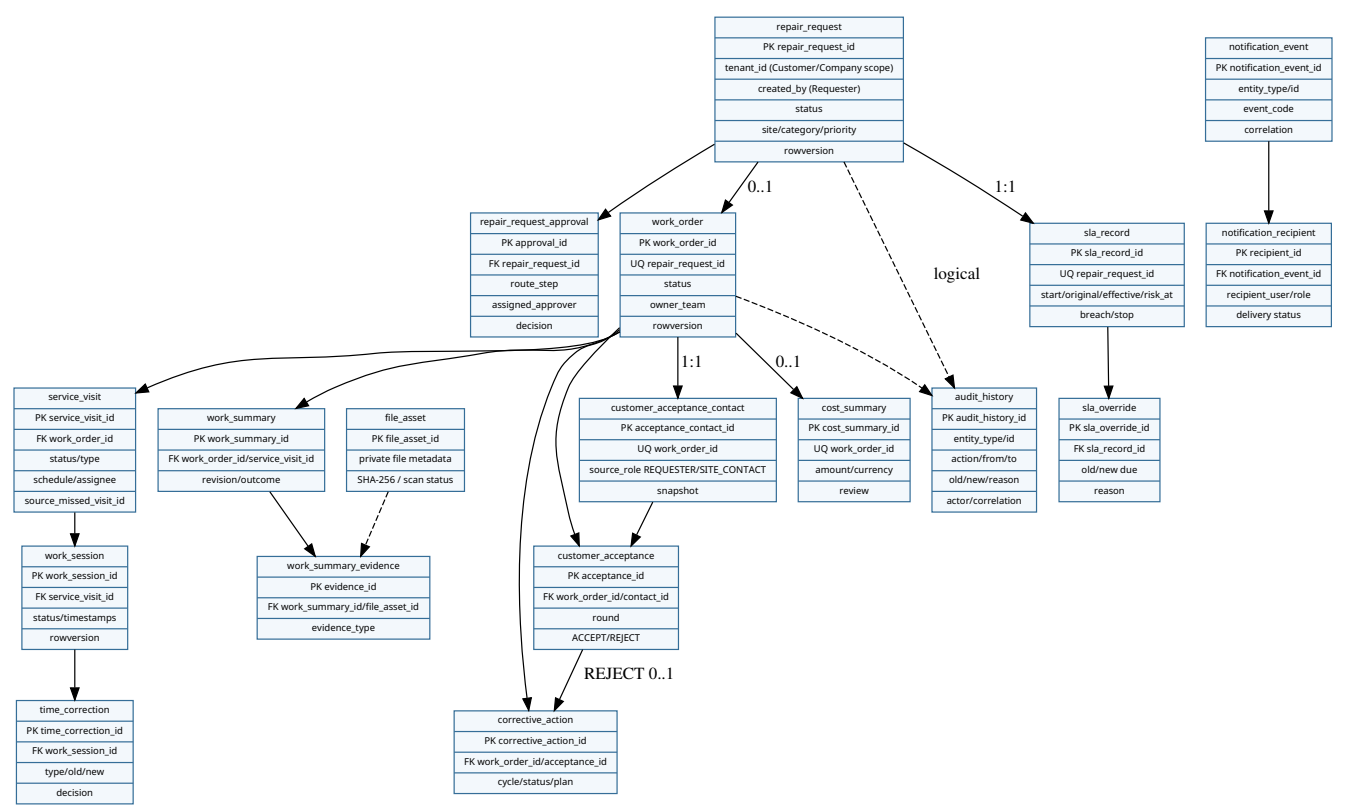


Figure 1 - Logical/physical entity baseline including NotificationRecipient and corrected acceptance contact semantics.

Repair Request

Field ID	Field	Datatype	Req.	Validation / Source / Audit
RR-001	repair_request_id	UUID	Y	PK; system generated
RR-002	tenant_id	UUID	Y	Customer/Company scope; server-derived; immutable
RR-003	request_no	varchar(30)	C	Unique; generated at SUBMITTED; immutable
RR-004	status	enum	Y	DRAFT/SUBMITTED/UNDER_REVIEW/APPROVED/REJECTED/CANCELLED/CONVERTED
RR-005	site_id	UUID	Y@Submit	Active Site; actor scope
RR-006	equipment_id	UUID	C	Active Equipment; belongs to Site; category rule
RR-007	location_id	UUID	C	Active Location; belongs to Site; category rule
RR-008	request_category_code	varchar(30)	Y@Submit	Active Category
RR-009	description	nvarchar(2000)	Y@Submit	Problem text; nonblank
RR-010	priority_code	varchar(20)	Y@Submit	Active Priority
RR-011	duplicate_continuation_reason	nvarchar(1000)	C	Required when BR-14 warning overridden
RR-012	created_by	UUID	Y	Requester identity; derived actor; immutable
RR-013	submitted_by	UUID	C	Derived actor on Submit
RR-014	submitted_at	datetime UTC	C	SLA start
RR-015	cancel_reason	nvarchar(1000)	C	Required Cancel
RR-016	reject_reason	nvarchar(1000)	C	Required Reject
RR-017	row_version	rowversion	Y	Optimistic concurrency
RR-018	request_contact_id	UUID	Y@Submit	Active contact in selected Site
RR-019	preferred_start_at	datetime UTC	Y@Submit	Preferred window start
RR-020	preferred_end_at	datetime UTC	Y@Submit	>= preferred_start_at

Approval Route / Decision

Field ID	Field	Datatype	Req.	Validation / Source / Audit
ARC-001	approval_route_id	UUID	Y	PK
ARC-002	tenant_id	UUID	Y	Scope
ARC-003	request_category_code	varchar(30)	Y	Active category mapping
ARC-004	site_id	UUID	C	null = tenant default
ARC-005	step_no	smallint	Y	Unique per route
ARC-006	approver_role_code	varchar(30)	Y	Authorized role
ARC-007	approver_user_id	UUID	C	Optional specific active approver
ARC-008	is_active	boolean	Y	Inactive route not selected
APR-001	approval_id	UUID	Y	PK
APR-003	repair_request_id	UUID	Y	FK Request

Field ID	Field	Datatype	Req.	Validation / Source / Audit
APR-004	approval_route_id	UUID	Y	FK Route
APR-005	approval_step_no	smallint	Y	Unique request/step
APR-006	assigned_approver_id	UUID	C	Missing -> routing failure
APR-007	status	enum	Y	PENDING/APPROVED/REJECTED/ RETURNED_FOR_CORRECTION (decision code)
APR-008	decision_reason	nvarchar(1000)	C	Required Reject/Return
APR-009	routed_at	datetime UTC	C	System time
APR-010	decided_at	datetime UTC	C	System time
APR-011	routing_failure_code	varchar(100)	C	Request remains SUBMITTED

File Asset / Attachment

Field ID	Field	Datatype	Req.	Validation / Source / Audit
FAS-001	file_asset_id	UUID	Y	PK shared private metadata
FAS-002	tenant_id	UUID	Y	Scope
FAS-003	file_name	nvarchar(255)	Y	Sanitized display
FAS-004	mime_type	varchar(100)	Y	PDF/JPG/JPEG/PNG
FAS-005	size_bytes	bigint	Y	<=10 MB
FAS-006	content_hash	char(64)	Y	SHA-256
FAS-007	storage_reference	nvarchar(500)	Y	Private object ref, never public URL
FAS-008	malware_scan_status	enum	Y	PENDING/CLEAN/FAILED
FAS-009	uploaded_by	UUID	Y	Derived actor
FAS-010	uploaded_at	datetime UTC	Y	Immutable
ATT-001	attachment_id	UUID	Y	PK
ATT-003	repair_request_id	UUID	Y	FK Request
ATT-013	file_asset_id	UUID	Y	FK FileAsset
ATT-014	access_scope_code	varchar(50)	Y	Role/site policy key

Work Order

Field ID	Field	Datatype	Req.	Validation / Source / Audit
WO-001	work_order_id	UUID	Y	PK
WO-002	tenant_id	UUID	Y	Scope
WO-003	work_order_no	varchar(30)	Y	Unique generated
WO-004	repair_request_id	UUID	Y	Unique FK; source APPROVED
WO-005	status	enum	Y	Canonical WO states only
WO-006	owner_team_id	UUID	C	Required before SCHEDULED
WO-007	team_lead_id	UUID	C	Active Team Lead

Field ID	Field	Datatype	Req.	Validation / Source / Audit
WO-008	acceptance_contact_id	UUID	Y@Before Acceptance	Exactly one designated contact
WO-009	acceptance_contact_snapshot	json	Y once set	Immutable display/contact snapshot
WO-010	cancel_reason	nvarchar(1000)	C	Required cancel
WO-011	closed_by	UUID	C	Supervisor
WO-012	closed_at	datetime UTC	C	On CLOSED
WO-013	row_version	rowversion	Y	Concurrency

Service Visit

Field ID	Field	Datatype	Req.	Validation / Source / Audit
SV-001	service_visit_id	UUID	Y	PK
SV-002	tenant_id	UUID	Y	Scope
SV-003	work_order_id	UUID	Y	FK Work Order
SV-004	visit_type	enum	Y	INITIAL/FOLLOW_UP/CORRECTIVE
SV-005	status	enum	Y	SCHEDULED/RESCHEDULED/IN_PROGRESS/COMPLETED/MISSED/CANCELLED
SV-006	assigned_team_id	UUID	C	Active authorized team
SV-007	assigned_technician_id	UUID	C	Active technician; site/team scope
SV-008	scheduled_start_at	datetime UTC	Y@Scheduled	<= end
SV-009	scheduled_end_at	datetime UTC	Y@Scheduled	>= start
SV-010	reschedule_reason	nvarchar(1000)	C	Required Reschedule
SV-011	reassign_reason	nvarchar(1000)	C	Required Reassign
SV-012	cancel_reason	nvarchar(1000)	C	Required pre-check-in Cancel
SV-013	missed_reason	nvarchar(1000)	C	Required MISSED
SV-014	completed_at	datetime UTC	C	Final Check-out
SV-015	row_version	rowversion	Y	Concurrency
SV-016	source_missed_visit_id	UUID	C	New visit link to original MISSED; original remains MISSED
SV-017	missed_decision_code	enum	C	RESCHEDULE/FOLLOW_UP/REASSIGN/NO_FOLLOW_UP
SV-018	missed_decided_at	datetime UTC	C	Decision timestamp

Work Session / Time Correction

Field ID	Field	Datatype	Req.	Validation / Source / Audit
WS-001	work_session_id	UUID	Y	PK
WS-003	service_visit_id	UUID	Y	FK Visit

Field ID	Field	Datatype	Req.	Validation / Source / Audit
WS-004	technician_id	UUID	Y	Derived/assigned actor
WS-005	status	enum	Y	CHECKED_IN/PAUSED/CHECKED_OUT
WS-006	check_in_at	datetime UTC	C	Corrections only via Time Correction
WS-007	pause_start_at	datetime UTC	C	One unmatched pause
WS-008	resume_at	datetime UTC	C	Requires active pause
WS-009	check_out_at	datetime UTC	C	Requires no active pause
WS-010	row_version	rowversion	Y	Concurrency
TC-001	time_correction_id	UUID	Y	PK
TC-002	work_session_id	UUID	Y	FK Session
TC-003	timestamp_type	enum	Y	CHECK_IN/PAUSE_START/RESUME/CHECK_OUT
TC-004	old_value	datetime UTC	Y	Immutable snapshot
TC-005	new_value	datetime UTC	Y	Must preserve chronology
TC-006	reason	nvarchar(1000)	Y	Nonblank
TC-007	requested_by	UUID	Y	Technician/Team Lead
TC-008	status	enum	Y	PENDING/APPROVED/REJECTED
TC-009	decided_by	UUID	C	Supervisor
TC-010	decision_reason	nvarchar(1000)	C	Required Reject
TC-011	decided_at	datetime UTC	C	System time
TC-012	sla_recalculated	boolean	Y	Always false MVP

Work Summary / Evidence / Cost

Field ID	Field	Datatype	Req.	Validation / Source / Audit
WSM-001	work_summary_id	UUID	Y	PK
WSM-003	work_order_id	UUID	Y	FK WO
WSM-004	service_visit_id	UUID	Y	FK Visit
WSM-005	revision_no	int	Y	Unique per Visit revision
WSM-006	summary_text	nvarchar(4000)	Y@Check-out	Nonblank
WSM-007	repair_outcome_code	varchar(30)	Y@Check-out	Active Outcome
EVD-001	evidence_id	UUID	Y	PK
EVD-003	work_summary_id	UUID	Y	FK Summary
EVD-004	evidence_type_code	varchar(30)	Y	Active Evidence Type
EVD-013	file_asset_id	UUID	Y	FK CLEAN FileAsset
CST-001	cost_summary_id	UUID	Y	PK
CST-003	work_order_id	UUID	Y	Unique current summary
CST-004	total_amount	decimal(18,2)		>=0

Field ID	Field	Datatype	Req.	Validation / Source / Audit
			Y@Before Close	
CST-005	currency_code	char(3)	Y	ISO currency
CST-006	note	nvarchar(2000)	C	Remark
CST-007	prepared_by	UUID	Y	Team Lead
CST-008	prepared_at	datetime UTC	Y	System time
CST-009	reviewed_by	UUID	C	Supervisor
CST-010	reviewed_at	datetime UTC	C	Required before CLOSED
CST-011	row_version	rowversion	Y	Concurrency

Acceptance / Corrective

Field ID	Field	Datatype	Req.	Validation / Source / Audit
CAC-001	acceptance_contact_id	UUID	Y	PK
CAC-003	work_order_id	UUID	Y	Unique 1:1
CAC-004	source_role	enum	Y	REQUESTER / SITE_CONTACT only
CAC-005	user_id	UUID	C	FK User when internal
CAC-006	full_name	nvarchar(200)	Y	Snapshot
CAC-007	email	varchar(254)	C	Restricted PII
CAC-008	phone	varchar(30)	C	Restricted PII
CAC-009	is_active	boolean	Y	Must be active at acceptance
CAC-010	designated_by	UUID	Y	Audit actor
ACC-001	acceptance_id	UUID	Y	PK
ACC-003	work_order_id	UUID	Y	FK WO
ACC-004	acceptance_round_no	smallint	Y	Unique per WO
ACC-005	acceptance_contact_id	UUID	Y	Must match designated
ACC-006	decision	enum	Y	ACCEPT/REJECT
ACC-007	decision_reason	nvarchar(1000)	C	Required REJECT
ACC-009	decided_at	datetime UTC	Y	System time
CA-001	corrective_action_id	UUID	Y	PK auto create on REJECT
CA-003	work_order_id	UUID	Y	FK WO
CA-004	acceptance_id	UUID	Y	FK rejected acceptance
CA-005	cycle_no	smallint	Y	Unique per WO
CA-006	status	enum	Y	DRAFT/PENDING_PLAN_APPROVAL/APPROVED
CA-007	owner_team_lead_id	UUID	Y	Team Lead
CA-008	plan_text	nvarchar(4000)	Y@Before Submit	Nonblank
CA-009	plan_file_asset_id	UUID	Y@Before Submit	CLEAN evidence

Field ID	Field	Datatype	Req.	Validation / Source / Audit
CA-010	approved_by	UUID	C	Supervisor
CA-011	approved_at	datetime UTC	C	System time
CA-012	corrective_service_visit_id	UUID	C	FK corrective Visit

SLA / Override

Field ID	Field	Datatype	Req.	Validation / Source / Audit
SLA-001	sla_record_id	UUID	Y	PK
SLA-003	repair_request_id	UUID	Y	Unique FK
SLA-004	sla_policy_version	varchar(30)	Y	Snapshot approved policy
SLA-005	started_at	datetime UTC	Y	SUBMITTED
SLA-006	original_due_at	datetime UTC	Y	started_at + 24h
SLA-007	effective_due_at	datetime UTC	Y	Default original; Override only
SLA-008	risk_at	datetime UTC	Y	effective_due_at - 4h
SLA-009	risk_notified_at	datetime UTC	C	Set when EV-SLA-002 emitted
SLA-010	breached_at	datetime UTC	C	Set once; sticky
SLA-011	stopped_at	datetime UTC	C	Defined terminal state
SLA-012	stop_reason	enum	C	REQUEST_REJECTED/REQUEST_CANCELLED/ WO_CLOSED/WO_CANCELLED
OVR-001	sla_override_id	UUID	Y	PK
OVR-002	sla_record_id	UUID	Y	FK
OVR-003	old_effective_due_at	datetime UTC	Y	Snapshot
OVR-004	new_effective_due_at	datetime UTC	Y	Authorized new due
OVR-005	reason	nvarchar(1000)	Y	Mandatory
OVR-006	overridden_by	UUID	Y	Supervisor with permission
OVR-007	overridden_at	datetime UTC	Y	System time
OVR-008	correlation_id	UUID	Y	Audit/outbox link

Notification / Audit

Field ID	Field	Datatype	Req.	Validation / Source / Audit
NTF-001	notification_event_id	UUID	Y	PK
NTF-002	tenant_id	UUID	Y	Scope
NTF-003	entity_type	varchar(30)	Y	Subject type
NTF-004	entity_id	UUID	Y	Subject id
NTF-005	event_code	varchar(50)	Y	Baseline NTF-* event
NTF-006	correlation_id	UUID	Y	Transition/outbox link
NTF-007	created_at	datetime UTC	Y	Event time

Field ID	Field	Datatype	Req.	Validation / Source / Audit
NTR-001	notification_recipient_id	UUID	Y	PK
NTR-002	notification_event_id	UUID	Y	FK event
NTR-003	recipient_user_id	UUID	C	Resolved specific recipient
NTR-004	recipient_role_code	varchar(30)	C	Routing source role
NTR-005	delivery_status	enum	Y	PENDING/SENT/FAILED
NTR-006	attempt_count	int	Y	Default 0
NTR-007	last_attempt_at	datetime UTC	C	Retry audit
AUD-001	audit_history_id	UUID	Y	PK append-only
AUD-003	entity_type	varchar(30)	Y	Subject
AUD-004	entity_id	UUID	Y	Subject id
AUD-005	action_code	varchar(50)	Y	Allowed action/transition
AUD-006	from_state	varchar(50)	C	Transition
AUD-007	to_state	varchar(50)	C	Transition
AUD-008	old_value_json	json	C	Field changes
AUD-009	new_value_json	json	C	Field changes
AUD-010	reason	nvarchar(1000)	C	Per rule
AUD-011	actor_id	UUID	Y	Derived actor/system
AUD-012	occurred_at	datetime UTC	Y	Immutable
AUD-013	correlation_id	UUID	Y	Trace

Validation / Security / Audit Controls

Control	Requirement
Allowed Action	Backend validates current From State, permission, tenant/site, ownership/assignment and row_version. UI/API cannot direct-write state.
Duplicate	D-12 active set + Site + Equipment/Location + Category + 24h; continuation reason if proceeding.
Conversion	Unique Request FK + Request CONVERTED + audit/outbox in one transaction; failure leaves APPROVED.
MISSED	Original visit remains MISSED; new visit references source_missed_visit_id; NO_FOLLOW_UP creates none.
Evidence	Category-driven master; corrective resubmission always CLEAN evidence.
SLA Risk	risk_at = effective_due_at - 4 hours; override recalculates risk_at only if not already breached/stopped; breach remains.
Notification	Event 1:N recipient rows; failed recipient delivery is retryable and does not alter business state.
Privacy	Contact/file content limited by tenant/role/site; no public URL or sensitive client log.

Resolved Configuration Decisions

Acceptance source_role is restricted to REQUESTER/SITE_CONTACT. OTHER_DESIGNATED is removed. Exact catalog values remain configuration; no additional business role is introduced.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อขัดแย้งกัน โดยไม่เปลี่ยน Business Baseline.

11. Data / Performance Consistency Corrections

11.1 Notification timestamp and index consistency

Canonical fields remain unchanged. `notification_event.created_at` is the event creation/order timestamp. `notification_recipient` contains recipient identity, delivery status, attempt count and `last_attempt_at`. This revision does **not** invent `notification_recipient.created_at`.

Any inbox/retry query or physical index must use fields that actually exist. If ordering by event time is required, the SQL query joins recipient to `notification_event` and the actual execution plan is verified. Denormalizing event time into recipient would be a separate Technical Schema Decision, not an implied baseline field.

11.2 Performance fixture vs data cardinality

“100,000 records per Site” is a Performance Test baseline dataset size, not a database cardinality constraint and not a mandatory production volume. Test fixtures may contain additional related WorkOrder, Visit, Audit and Notification rows to exercise representative joins.

11.3 Concurrency field rule

Existing `row_version` / SQL rowversion fields remain the concurrency authority on mutable aggregates defined by the baseline. Performance tuning may not remove or bypass the token on state-changing/update commands.

12. Data Evidence Required for Performance Test

Evidence	Purpose
Dataset seed/version + row counts by key entity	Repeatability and data-shape traceability
Generated SQL sample for critical read paths	Verify server-side predicates/projection and absence of accidental client filtering
Actual plan / Query Store or equivalent evidence	Validate candidate index strategy against real access pattern
Audit/notification row growth snapshot	Detect timeline/inbox/worker path scaling risk